ENTERPRISE SYSTEM BLUEPRINT & DATA COLLECTION SPECIFICATION

Standard Master Data Formats, Process Flow Charts & Module Dashboards

A comprehensive operational manual for department heads (Sales, Engineering, Purchase, Quality, Store, Production, Dispatch, and After Sales) defining the exact input templates, sequential stage gates, and real-time plant KPIs required for the Machine House ETO manufacturing execution platform.

1. Standardized Master Data

Clean Excel schemas with column names, data types, mandatory flags, validation constraints, and realistic industrial winding machine values.

2. Visual Process Flow Charts

End-to-end operational sequence from lead capture to site SAT, including automated ECO freeze rules, hard allocation locks, and zero-snag stops.

3. Live Module Dashboards

Executive and shopfloor dashboards featuring 4 top-level KPI metrics, active priority queues, and real-time distribution charts.

Corporate Entity: Machine House - Winding Technologies Pvt Ltd
Manufacturing Plants: Unit 1 Bhosari (Pune) & Unit 2 Chakan Industrial Area

Version: Release 1.0 (Enterprise ETO Edition)
Publish Date: October 2026 | Confidential Internal Plant Blueprint

SPINE ETO Manufacturing Architecture & Central Project Spine

Machine House Operations Core

In an Engineer-to-Order (ETO) plant, every machine is engineered to specific client coil specifications. The **Unique Project ID / Master Work Order ID (PRJ-XXXX / WO-XXXX)** acts as the central spine connecting all downstream procurement, quality inspections, store inventory allocations, shopfloor assembly stages, dry/wet trials, and zero-snag dispatches.



COMMON FOUNDATION MASTERS (SHARED ENTERPRISE LOOKUPS)

Company Profile Master (company_profile.xlsx) — Legal and tax configuration for all documents, invoices, delivery challans, and test certs.

Field Name (Excel Column)	Data Type	Mandatory?	Validation / Constraint	Sample Industrial Value
`legalEntityName`	String	Mandatory	Exact registered name	**Machine House - Winding Technologies Pvt Ltd**
`tagline`	String	Optional	Corporate slogan	**House of Innovations.....**
`cin`	String	Mandatory	21-character CIN	**U29253PN2015PTC155421**
`pan`	String	Mandatory	10-character PAN	**AACCM8892K**
`gstin`	String	Mandatory	15-character GSTIN	**27AACCM8892K1Z8**
`registeredOffice`	String	Mandatory	Full registered address with PIN	**Plot No. 44, Electronic Zone, MIDC Bhosari, Pune - 411026**
`plantUnit1`	String	Mandatory	Plant address	**Plot No. 12/B, Industrial Sector 7, Chakan Industrial Area, Pune - 410501**

Item Category & Taxonomy Master (categories.xlsx) — Taxonomy governing inventory segregation, store indexing, and QC requirements.

Field Name (Excel Column)	Data Type	Mandatory?	Validation / Constraint	Sample Industrial Value
`categoryCode`	String	Mandatory	RAW_MAT / BO_ELEC / BO_MECH / BO_PNU / MACH_PART / SUB_ASSY / FIN_MACH	**BO_ELEC**
`categoryName`	String	Mandatory	Descriptive category title	**Bought-Out Electrical & Automation**
`classificationType`	Dropdown	Mandatory	RM / BO / COMPONENT / SUBASSEMBLY / ASSEMBLY	**BO**
`requiresQc`	Boolean	Mandatory	True enforces gate quarantine inspection	**TRUE**

Unit of Measure (UOM) & Dual-Unit Normalization (uoms.xlsx) — Governs discrete vs continuous units and locks mathematical conversion factors (e.g. Kg vs Meters).

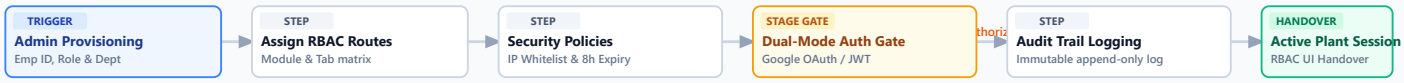
Field Name (Excel Column)	Data Type	Mandatory?	Validation / Constraint	Sample Industrial Value
`uomCode`	String	Mandatory	NOS / SET / KG / MTR / LTR / BOX	**KG**
`uomName`	String	Mandatory	Full metric name	**Kilograms**
`metricType`	Dropdown	Mandatory	DISCRETE (Integer only) / CONTINUOUS (Fractional allowed)	**CONTINUOUS**
`conversionFormula`	String	Optional	Formula for alternate UOM (e.g. 1 Box = 50 NOS)	**weight = length * unitWeight**

M1 System Admin & Governance

Route: `/admin` | Owner: **IT & Managing Director**

Central control center for user identity, URL-route RBAC permissions matrix, plant IP whitelisting, and immutable audit logs.

1. VISUAL PROCESS FLOW CHART & STAGE GATES



2. MODULE DASHBOARD & REAL-TIME KPIS (WHAT PLANT USERS SEE)

System Admin & Governance Dashboard (Live Plant View) — Real-time sync active

ACTIVE PLANT SESSIONS	SECURITY AUDITS (24H)	FAILED LOGINS LOCKED	PLANT IP COMPLIANCE
24 Active	**0 High Risk**	**2 Blocked**	**100%**
Normal	100% Clean	5-Attempt Rule	MIDC Whitelist

Live System Audit Trail

Timestamp	User Account	Department	Action Executed	Target Entity	IP Address	Status
10:04:12	rajesh.k@machinehouse.in	QUALITY	STAGE_GATE_PASS	PRJ-2026-001 (PDI Sign-off)	192.168.1.45 (Bhosari)	SUCCESS
09:48:30	arshad@machinehouse.in	MANAGEMENT	ECO_APPROVAL	ECO-2026-004 (Spindle Shaft)	192.168.1.10 (Office)	SUCCESS
09:22:15	amit.p@machinehouse.in	PURCHASE	PO_RELEASE	PO-2026-0142 (Siemens Motors)	192.168.1.52 (Chakan)	SUCCESS
08:50:00	unknown_guest@gmail.com	EXTERNAL	UNAUTHORIZED_LOGIN	Authentication Route	103.22.44.12 (External)	BLOCKED

Departmental User Distribution

Production & Assembly	8 Users
Quality Assurance	5 Users
Design & Engineering	4 Users
Store & Logistics	4 Users
Purchase & Accounts	3 Users

3. MASTER DATA REQUIREMENTS & EXCEL INPUT SCHEMAS

User Master (users.xlsx)

Field Name (Excel Header)	Data Type	Mandatory?	Validation Rules / Constraints	Sample Industrial Value
employeeId	String	Mandatory	Unique ID format MHT-EMP-XXX	**MHT-EMP-014**
fullName	String	Mandatory	Full legal name	**Rajesh Kulkarni**
email	String	Mandatory	Official corporate email domain	**rajesh.k@machinehouse.in**
department	Dropdown	Mandatory	DESIGN / PURCHASE / QUALITY / STORE / PRODUCTION	**QUALITY**
primaryRole	Dropdown	Mandatory	References Role Master	**QC_LEAD**
sessionRefreshExpiry	Dropdown	Mandatory	8_HOURS (Kiosk) / 7_DAYS / 30_DAYS	**8_HOURS**
ipRestricted	Boolean	Mandatory	True to restrict login to Plant IP whitelist	**TRUE**

📁 Role Route RBAC Matrix (roles.xlsx)

Field Name (Excel Header)	Data Type	Mandatory?	Validation Rules / Constraints	Sample Industrial Value
`roleCode`	String	Mandatory	Unique Role Identifier	**QC_LEAD**
`roleName`	String	Mandatory	Descriptive Role Title	**Quality Assurance Lead**
`department`	Dropdown	Mandatory	Target operational department	**QUALITY**
`permittedTabRoutes`	List	Mandatory	Granular URL routes (e.g. /quality/inward, /quality/pdi)	**/quality/inward, /quality/pdi**

M2 CRM, Enquiry & Costing

Route: `/crm` | Owner: **Sales & Business Development**

Lead capture, Technical Requirement Specification (TRS), engineering costing estimation, and 1-click transition to firm project.

1. VISUAL PROCESS FLOW CHART & STAGE GATES



2. MODULE DASHBOARD & REAL-TIME KPIS (WHAT PLANT USERS SEE)

CRM, Enquiry & Costing Dashboard (Live Plant View) — Real-time sync active

ACTIVE PIPELINE VALUE	QUOTATION WIN RATIO	AVG QUOTE TURNAROUND	ADVANCE COLLECTED
₹ 4.82 Cr	**42.5%**	**3.2 Days**	**₹ 85.0 L**
18 Enquiries	+3.2% vs target	SLA < 4 Days	100% Cleared

High-Priority Quotations & Enquiry Queue

Enquiry ID	Customer Name	Machine Classification	Target Delivery	Estimated Value	Payment Terms	Stage Status
ENQ-2026-089	ABB Power Products Vadodara	4-Spindle High Speed Coil Winder	8 Weeks	₹ 18,50,000	30% Adv / 60% PI / 10% SAT	WON - CONVERTING
ENQ-2026-092	BHEL Bhopal	5-Axis CNC Precision Gantry	12 Weeks	₹ 42,00,000	20% Adv / 70% PI / 10% SAT	QUOTE_SUBMITTED
ENQ-2026-094	Schneider Electric Hyderabad	Automated Wire Tensioner Test Bench	6 Weeks	₹ 9,20,000	50% Adv / 40% PI / 10% SAT	COSTING_REVIEW
ENQ-2026-098	Lucas TVS Chennai	Dual-Spindle Stator Coil Winder	10 Weeks	₹ 24,00,000	Standard Terms	TRS_DRAFT

Enquiry Conversion Funnel

New Leads Received	18 Leads
TRS Technical Feasibility	14 Qualified
Costing & Commercial Quote	10 Sent
Customer PO Won (Firm)	4 Confirmed

3. MASTER DATA REQUIREMENTS & EXCEL INPUT SCHEMAS

Customer Master (customers.xlsx)

Field Name (Excel Header)	Data Type	Mandatory?	Validation Rules / Constraints	Sample Industrial Value
`customerCode`	String	Mandatory	Unique ID format CUST-XXX	**CUST-001**
`companyName`	String	Mandatory	Registered corporate name	**ABB Power Products & Systems Ltd**
`gstin`	String	Mandatory	15-character valid Indian GSTIN	**24AABCA1122K1Z5**
`billingAddress`	String	Mandatory	Registered billing address with PIN	**Maneja Works, Vadodara - 390013**
`contactPerson`	String	Mandatory	Procurement officer name	**Rajesh Mehta**
`contactEmail`	String	Mandatory	Official procurement email	**procure@abb.com**
`paymentTerms`	Dropdown	Mandatory	Commercial milestone terms	**30% Advance, 60% PI, 10% SAT**

📁 Lead & TRS Specification Master (leads.xlsx)

Field Name (Excel Header)	Data Type	Mandatory?	Validation Rules / Constraints	Sample Industrial Value
`leadCode`	String	Mandatory	Unique Enquiry ID format ENQ-YYYY-XXX	**ENQ-2026-089**
`customerCode`	Dropdown	Mandatory	References Customer Master	**CUST-001**
`targetMachineType`	Dropdown	Mandatory	Coil Winder / CNC Gantry / Tensioner	**4-Spindle High Speed Coil Winder**
`wireGaugeRange`	String	Mandatory	Wire technical range specification	**0.15 mm to 2.5 mm Copper Wire**

M3 Project Setup — The Central Spine

Route: `/project-setup` | Owner: **Project Manager & Operations Head**

Establishes the permanent lifecycle reference spine (PRJ-XXXX / WO-XXXX) governing all downstream engineering, purchasing, and trials.

1. VISUAL PROCESS FLOW CHART & STAGE GATES



2. MODULE DASHBOARD & REAL-TIME KPIS (WHAT PLANT USERS SEE)

Project Setup — The Central Spine Dashboard (Live Plant View) — Real-time sync active

ACTIVE PLANT PROJECTS	ON-TIME MILESTONE INDEX	CRITICAL PATH BOTTLENECKS	AVERAGE ETO LEAD TIME
12 Machines	**91.6%**	**1 Delayed**	**9.4 Weeks**
Bhosari & Chakan	+2.4% vs target	Needs Attention	Industry Benchmark

Active Project Execution Spine (Master Work Orders)

Project ID	Work Order	Customer Name	Machine Classification	Current Stage Gate	Committed Delivery	Overall Health
PRJ-2026-001	WO-2026-001	ABB Power Products	4-Spindle High Speed Coil Winder	SUB_ASSY (Sub-assembly Trial)	2026-05-15	ON_TRACK
PRJ-2026-002	WO-2026-002	BHEL Bhopal	5-Axis CNC Coil Forming Center	PROCURE (PO Placement)	2026-06-30	ON_TRACK
PRJ-2026-003	WO-2026-003	Schneider Electric	Rotary Turret Indexing Station	DESIGN_BOM (Drawing Release)	2026-05-05	DELAY_RISK
PRJ-2026-004	WO-2026-004	Lucas TVS	Dual Spindle Automatic Winder	TRIAL_RUN (Dry Run Testing)	2026-04-20	ON_TRACK

Plant Machine Stage Distribution

Stage 2: Design & CAD Modeling	2 Machines
Stage 3: Procurement & Inward QC	3 Machines
Stage 4: Sub-Assembly & Panel Wiring	4 Machines
Stage 5: Machine Trials & Testing	2 Machines
Stage 6: Zero-Snag Dispatch	1 Machine

3. MASTER DATA REQUIREMENTS & EXCEL INPUT SCHEMAS

Project Master (projects.xlsx)

Field Name (Excel Header)	Data Type	Mandatory?	Validation Rules / Constraints	Sample Industrial Value
projectId	String	Mandatory	Unique ID format PRJ-YYYY-XXX	**PRJ-2026-001**
workOrderNumber	String	Mandatory	Plant WO format WO-YYYY-XXX	**WO-2026-001**
customerCode	Dropdown	Mandatory	References Customer Master	**CUST-001 (ABB)**
machineModel	String	Mandatory	Full machine model name	**Automated 4-Spindle High Speed Coil Winder**
contractedDeliveryDate	Date	Mandatory	Committed dispatch date	**2026-05-15**
projectManager	Dropdown	Mandatory	Assigned internal project head	**Rajesh Kulkarni**

📁 Standard Stage Master (stages.xlsx)

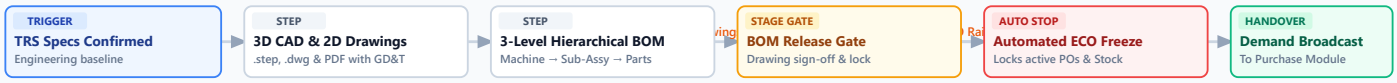
Field Name (Excel Header)	Data Type	Mandatory?	Validation Rules / Constraints	Sample Industrial Value
`stageSequence`	Number	Mandatory	Sequence 1 to 10	**1, 2, 3... 10**
`stageCode`	String	Mandatory	KICKOFF / DESIGN_BOM / PROCURE / SUB_ASSY / TRIAL	**DESIGN_BOM**
`mandatorySignoffRole`	Dropdown	Mandatory	Role authorized to approve stage	**DES_ENG**

M4 Engineering, CAD & Hierarchical BOM

Route: /design | Owner: **Head of Design & Engineering**

Constructs 3-level hierarchical BOMs, manages CAD/drawing distribution, and enforces automated ECO freeze rules.

1. VISUAL PROCESS FLOW CHART & STAGE GATES



2. MODULE DASHBOARD & REAL-TIME KPIS (WHAT PLANT USERS SEE)

Engineering, CAD & Hierarchical BOM Dashboard (Live Plant View) — Real-time sync active

BOM RELEASE PROGRESS	ACTIVE OPEN ECOS	AVG DRAWING REVISIONS	STANDARD BO RATIO
98.4%	**2 Open**	**1.4 / Part**	**68%**
122/124 Parts Released	PO Lines Frozen	Within ISO Tolerance	High Standardization

3-Level Hierarchical BOM Explorer (PRJ-2026-001)

BOM Level	Item Code	Item Description	Category	Qty/Mch	UOM	Drawing / Specification No	Status
Level 0	ASY-CW-400	4-Spindle High Speed Coil Winder	FIN_MACH	1	NOS	GA-MHT-CW400-00	RELEASED
Level 1	SA-WND-HEAD	High-Torque Precision Winding Head Assembly	SUB_ASSY	4	SET	SUB-MHT-WH-01	RELEASED
Level 2	CMP-SFT-001	Main Spindle Shaft Hardened & Ground	MACH_PART	4	NOS	DWG-MHT-001-R3	RELEASED
Level 2	BO-SRV-750W	750W AC Servo Motor with Brake (Siemens)	BO_ELEC	4	SET	1FK7042-2AF71-1QA0	RELEASED
Level 2	BO-BRG-6205	Deep Groove Ball Bearing 25mm ID (SKF)	BO_MECH	8	NOS	SKF 6205-2RSH	RELEASED
Level 1	SA-TNS-UNIT	Closed-Loop Servo Wire Tensioner Unit	SUB_ASSY	4	SET	SUB-MHT-TU-02	FROZEN_ECO

Engineering Change Order (ECO) Impact

ECO-2026-004: Spindle Bearing Tolerance	PO-0142 Frozen
ECO-2026-005: Tensioner Arm Bracket	Stock Reserved
Drawing Watermark Protection	100% Active
CAD File Whitelist (.step / .dwg / .pdf)	Enforced

3. MASTER DATA REQUIREMENTS & EXCEL INPUT SCHEMAS

3-Level BOM Master (bom_template.xlsx)

Field Name (Excel Header)	Data Type	Mandatory?	Validation Rules / Constraints	Sample Industrial Value
bomLevel	Dropdown	Mandatory	0 (Machine) / 1 (Sub-Assembly) / 2 (Part)	**Level 1**
parentPartCode	String	Optional	Parent Assembly code (Blank for Level 0)	**ASY-CW-400**
itemCode	Dropdown	Mandatory	References Component / BO Master	**SA-WND-HEAD**
qtyPerMachine	Number	Mandatory	Quantity required per machine	**4**
uom	Dropdown	Mandatory	NOS / SET / KG / MTR	**SET**

📁 Drawing & CAD Document Master (drawings.xlsx)

Field Name (Excel Header)	Data Type	Mandatory?	Validation Rules / Constraints	Sample Industrial Value
`drawingNumber`	String	Mandatory	Standard format DWG-MHT-XXX-RX	**DWG-MHT-001-R3**
`drawingRevision`	String	Mandatory	Revision tag R0, R1, R2...	**R3**
`cadFileType`	Dropdown	Mandatory	.step / .dwg / .dxf / .pdf	**.step / .pdf**

M5 Purchase, Vendor Sourcing & Demands

Route: /purchase | Owner: **Head of Purchase & Sourcing**

Calculates BOM net purchase demand (Gross BOM minus Free Store Stock), manages vendor approvals, and tracks overdue deliveries.

1. VISUAL PROCESS FLOW CHART & STAGE GATES



2. MODULE DASHBOARD & REAL-TIME KPIS (WHAT PLANT USERS SEE)

Purchase, Vendor Sourcing & Demands Dashboard (Live Plant View) — Real-time sync active

ACTIVE COMMITTED PO SPEND	OVERDUE PO DELIVERIES	VENDOR OTIF SCORE	BOM NET DEMAND COVERED
₹ 62.4 Lakhs	**3 POs**	**94.2%**	**98.0%**
28 Active POs	Expediting Active	+1.8% Quality Pass	120/124 Parts Covered

Vendor Procurement & Overdue Delivery Queue

PO Number	Project Spine	Supplier Name	Item Description	Order Qty	Delivery Due Date	Overdue Aging	PO Status
PO-2026-0142	PRJ-2026-001	Siemens Automation India	750W Servo Motor with Brake	4 SET	2026-03-23	3 Days Overdue	FOLLOWUP_ACTIVE
PO-2026-0145	PRJ-2026-001	SKF India Bearings Ltd	Deep Groove Ball Bearing 6205	8 NOS	2026-03-20	5 Days Overdue	DISPATCHED_IN_TRANSIT
PO-2026-0150	PRJ-2026-002	Hiwin Precision Motion	Linear Guide Rails 25x1200mm	12 SET	2026-04-05	On-Schedule	APPROVED_SENT
PO-2026-0154	PRJ-2026-003	Festo Pneumatics Pvt Ltd	5/2 Way Solenoid Valves 24VDC	6 NOS	2026-04-12	On-Schedule	PENDING_HOD_APPROVAL

Procurement Spend by Category

Bought-Out Electrical & Drives	₹ 24.8 L
Bought-Out Mechanical & Motion	₹ 18.2 L
Raw Material Steels & Alloys	₹ 11.5 L
Pneumatics & Fluid Power	₹ 7.9 L

3. MASTER DATA REQUIREMENTS & EXCEL INPUT SCHEMAS

Vendor Master (vendors.xlsx)

Field Name (Excel Header)	Data Type	Mandatory?	Validation Rules / Constraints	Sample Industrial Value
vendorCode	String	Mandatory	Unique ID format VND-XXX	**VND-001**
vendorName	String	Mandatory	Registered supplier name	**Siemens Automation India Ltd**
category	Dropdown	Mandatory	ELECTRICAL / MECHANICAL / PNEUMATICS / RM	**ELECTRICAL**
ratingScore	Number	Mandatory	1.0 to 5.0 OTIF rating	**4.8**
gstin	String	Mandatory	15-character valid GSTIN	**27AABCS1429B1Z1**
creditPeriodDays	Number	Mandatory	Agreed credit terms in days	**45**

📁 Item Purchase Specs Master (item_specs.xlsx)

Field Name (Excel Header)	Data Type	Mandatory?	Validation Rules / Constraints	Sample Industrial Value
`itemCode`	Dropdown	Mandatory	References Component / BO Master	**BO-SRV-750W**
`preferredVendor`	Dropdown	Mandatory	References Vendor Master	**VND-001 (Siemens)**
`standardLeadDays`	Number	Mandatory	Lead time in days	**21**
`hsnCode`	String	Mandatory	8-digit HSN code	**85015290**

M6 Quality Assurance & Inspections

Route: `/quality` | Owner: **Head of Quality Assurance**

3-Tier QC architecture: Inward QC (quarantine hold), Assembly In-line Stage Inspections, and Pre-Delivery Inspection (PDI).

1. VISUAL PROCESS FLOW CHART & STAGE GATES



2. MODULE DASHBOARD & REAL-TIME KPIS (WHAT PLANT USERS SEE)

Quality Assurance & Inspections Dashboard (Live Plant View) — Real-time sync active

INWARD QC TURNAROUND	FIRST TIME PASS RATE (FTPR)	QUARANTINE HOLDING LOTS	PDI ZERO-SNAG INDEX
4.8 Hours	**96.8%**	**3 Lots**	**85.0%**
SLA < 8 Hours	+1.2% this month	Awaiting Inspection	Pre-Dispatch First Pass

Pending Inward QC Quarantine Queue (Holding Bay)

Gate Entry #	Vendor Supplier	Item Description	Challan Qty	Arrival Timestamp	Aging in Bay	Inspection Status
GE-2026-0312	Siemens Automation India	750W AC Servo Motor with Brake	4 SET	Today 08:30 AM	3.5 Hours	INSPECTION_IN_PROGRESS
GE-2026-0314	Festo Pneumatics Pvt Ltd	Micro Cylinders DSNU-16-50	4 NOS	Today 09:15 AM	2.8 Hours	PENDING_MTC_VERIFICATION
GE-2026-0318	Precision CNC Job Works	Main Spindle Shaft Ground	4 NOS	Today 10:00 AM	2.0 Hours	QUEUED_FOR_MICROMETER
GE-2026-0305	Alloy Steel Corporation	EN19 Alloy Steel Round Bar 75mm	220 KG	Yesterday 04:30 PM	18.5 Hours	ACCEPTED_CLEARED_TO_STORE

Supplier Quality Defect Causes (Pareto)

Dimensional Deviation (OD / Bore)	45%
Surface Finish Roughness > Ra limit	25%
Missing Material Test Certificate (MTC)	15%
Core Hardness Below Spec Range	15%

3. MASTER DATA REQUIREMENTS & EXCEL INPUT SCHEMAS

Quality Inspection Parameter Master (qc_parameters.xlsx)

Field Name (Excel Header)	Data Type	Mandatory?	Validation Rules / Constraints	Sample Industrial Value
`parameterCode`	String	Mandatory	Unique ID format QC-PAR-XXX	**QC-PAR-012**
`parameterName`	String	Mandatory	Inspection attribute name	**Spindle Bearing Journal Diameter**
`inspectionMethod`	Dropdown	Mandatory	VERNIER / MICROMETER / CMM / HARDNESS	**MICROMETER**
`nominalValue`	Number	Mandatory	Target dimension	**25.000**
`upperTolerance`	Number	Mandatory	Positive tolerance	**+0.009**
`lowerTolerance`	Number	Mandatory	Negative tolerance	**-0.000**

📁 Defect Code Master (defect_codes.xlsx)

Field Name (Excel Header)	Data Type	Mandatory?	Validation Rules / Constraints	Sample Industrial Value
`defectCode`	String	Mandatory	Format DEF-XXX-XXX	**DEF-DIM-US**
`defectDescription`	String	Mandatory	Description of defect	**Dimension Out of Spec (Undersized)**
`severityLevel`	Dropdown	Mandatory	CRITICAL / MAJOR / MINOR	**CRITICAL**

M7 Store, Inventory & Hard Allocations

Route: `/store` | Owner: **Store In-charge & Logistics Lead**

Tracks 4 inventory states (Ordered, In-QC, Accepted, Issued) and locks materials to target project IDs to block unauthorized diversion.

1. VISUAL PROCESS FLOW CHART & STAGE GATES



2. MODULE DASHBOARD & REAL-TIME KPIS (WHAT PLANT USERS SEE)

Store, Inventory & Hard Allocations Dashboard (Live Plant View) — Real-time sync active

TOTAL INVENTORY VALUATION	HARD-ALLOCATED PROPORTION	QUARANTINE HOLDING VALUE	MISSING PARTS / STOCKOUTS
₹ 1.42 Cr	**82.5%**	**₹ 6.8 Lakhs**	**0 Parts**
Accepted Stock	Committed to Active PRJs	3 Lots in IQC	Assembly Kits Ready

4-Bucket Inventory Ledger & Project Allocations

Item Code	Item Description	Category	Warehouse Bin	Ordered	In-QC	Accepted	Issued	Allocation Status
BO-SRV-750W	750W AC Servo Motor with Brake	BO_ELEC	BIN-B205 (Aisle B)	4 SET	4 SET	8 SET	4 SET	HARD_LOCKED (PRJ-2026-001)
BO-GBX-010	Planetary Gearbox 10:1 Ratio	BO_MECH	BIN-B206 (Aisle B)	2 NOS	0 NOS	4 NOS	2 NOS	HARD_LOCKED (PRJ-2026-001)
BO-LNR-25	Linear Rail Guideway 25x1200mm	BO_MECH	BIN-B301 (Aisle B)	12 SET	0 SET	12 SET	6 SET	HARD_LOCKED (PRJ-2026-002)
RM-EN19-075	EN19 Alloy Steel Round Bar 75mm	RAW_MAT	BIN-A101 (Aisle A)	0 KG	0 KG	220 KG	150 KG	UNRESERVED (Free Stock 70 KG)

Project Kitting Readiness Progress

PRJ-2026-001 (ABB 4-Spindle Winder)	94% Kitted
PRJ-2026-002 (BHEL 5-Axis Gantry)	82% Kitted
PRJ-2026-003 (Schneider Turret)	65% Kitted
PRJ-2026-004 (Lucas TVS Dual Winder)	100% Kitted

3. MASTER DATA REQUIREMENTS & EXCEL INPUT SCHEMAS

Warehouse Location & Bin Master (locations.xlsx)

Field Name (Excel Header)	Data Type	Mandatory?	Validation Rules / Constraints	Sample Industrial Value
locationCode	String	Mandatory	Unique ID format LOC-XXX	**LOC-002**
plantName	Dropdown	Mandatory	Unit 1 Bhosari / Unit 2 Chakan	**Unit 1 Bhosari**
warehouseType	Dropdown	Mandatory	RAW_MATERIAL / BOUGHT_OUT / QUARANTINE	**BOUGHT_OUT**
rackNumber	String	Mandatory	Racking identifier	**R-04**
binNumber	String	Mandatory	Physical bin barcode identifier	**BIN-B205**

📁 Project Hard Allocation Ledger (stock_allocations.xlsx)

Field Name (Excel Header)	Data Type	Mandatory?	Validation Rules / Constraints	Sample Industrial Value
`allocationId`	String	Mandatory	Unique ID format ALC-XXXXX	**ALC-00142**
`projectId`	Dropdown	Mandatory	Target Project Spine PRJ-XXXX	**PRJ-2026-001**
`itemCode`	Dropdown	Mandatory	References Component / BO Master	**BO-SRV-750W**
`allocatedQty`	Number	Mandatory	Hard reserved quantity	**4**
`allocationStatus`	Dropdown	Mandatory	LOCKED / TRANSFERRED / ISSUED	**LOCKED**

M8 Production, Work Centers & Machine Trials

Route: `/production-trial` | Owner: **Head of Production & Assembly**

Configures assembly bay routings, stage gate sign-offs, and continuous 4-hour machine trials with snag rectification tracking.

1. VISUAL PROCESS FLOW CHART & STAGE GATES



2. MODULE DASHBOARD & REAL-TIME KPIS (WHAT PLANT USERS SEE)

Production, Work Centers & Machine Trials Dashboard (Live Plant View) — Real-time sync active

TRIAL FIRST PASS YIELD	ACTIVE OPEN SNAGS	AVG SNAG CLEARANCE TIME	ASSEMBLY SCHEDULE ADHERENCE
88.5%	**4 Snags**	**6.2 Hours**	**94.0%**
+3.5% vs baseline	All Under Rectification	SLA < 8 Hours	On Schedule

Shopfloor Work Center & Machine Test Stand Matrix

Work Center Bay	Dedicated Function	Active Machine	Stage In-Progress	Assigned Foreman	Running Hours	Status
`WC-BAY-01 (Bhosari)`	Main Frame & Gantry Assembly	PRJ-2026-003	Base Frame Alignment	Santosh Shinde	18 Hours	ACTIVE_ASSEMBLY
`WC-BAY-02 (Bhosari)`	Precision Sub-Assembly Bay	PRJ-2026-002	Servo Tensioner Assembly	Prakash Patil	24 Hours	ACTIVE_ASSEMBLY
`WC-BAY-03 (Bhosari)`	Electrical & Control Panels	PRJ-2026-001	Panel Wire Dressing	Nitin More	36 Hours	ACTIVE_WIRING
`WC-BAY-04 (Chakan)`	Machine Trial & Testing Stand	PRJ-2026-004	Continuous 4h Dry Run	Suresh Jadhav	3.2 / 4.0 Hours	TESTING_IN_PROGRESS

Active Assembly Snags by Severity

Critical Snags (Block Trials)	0 Critical
Major Snags (Functional Rectification)	2 Major
Minor Snags (Cosmetic / Wire Tidy)	2 Minor
Snag Clearance Turnaround SLA	6.2 Hours

3. MASTER DATA REQUIREMENTS & EXCEL INPUT SCHEMAS

Work Center & Bay Master (work_centers.xlsx)

Field Name (Excel Header)	Data Type	Mandatory?	Validation Rules / Constraints	Sample Industrial Value
`workCenterCode`	String	Mandatory	Unique ID format WC-BAY-XX	**WC-BAY-04**
`bayName`	String	Mandatory	Descriptive bay title	**Machine Trial & Testing Stand**
`plantLocation`	Dropdown	Mandatory	Unit 1 Bhosari / Unit 2 Chakan	**Unit 2 Chakan**
`capacity`	String	Mandatory	Max machines / units in bay	**2 Machine Test Stands**

📁 Assembly Stage Protocol Master (assembly_stages.xlsx)

Field Name (Excel Header)	Data Type	Mandatory?	Validation Rules / Constraints	Sample Industrial Value
`stageSequence`	String	Mandatory	Sequence ID format STG-XX	**STG-60**
`stageName`	String	Mandatory	Stage title	**Machine Dry Run (Continuous 4 Hours)**
`targetWorkCenter`	Dropdown	Mandatory	References Work Center Master	**WC-BAY-04**
`standardManHours`	Number	Mandatory	Planned execution hours	**12**

M9 Dispatch, Packing & Logistics

Route: `/dispatch` | Owner: **Plant Head & Dispatch Manager**

Enforces the absolute zero-snag rule (100% snags resolved before gate pass unlocks), packing crate lists, and shipping dossier verification.

1. VISUAL PROCESS FLOW CHART & STAGE GATES



2. MODULE DASHBOARD & REAL-TIME KPIS (WHAT PLANT USERS SEE)

Dispatch, Packing & Logistics Dashboard (Live Plant View) — Real-time sync active

ON-TIME PLANT DISPATCH	READY-TO-SHIP MACHINES	ZERO-SNAG GATE PASS PASS	AVERAGE TRANSIT LEAD TIME
93.8%	**2 Machines**	**100%**	**2.4 Days**
+1.8% vs benchmark	Crated & Sealed	Zero Exceptions Rule	V-Trans Logistics

Ready for Dispatch Checklist & Pre-Dispatch Gate Pass Queue

Project ID	Customer Company	Machine Model	Crates Count	PDI Sign-off	Commercial Clearance	Gate Pass Status
PRJ-2026-004	Lucas TVS Chennai	Dual Spindle Automatic Winder	3 Wooden Crates	SIGNED (100% PASS)	CLEARED (100% Invoiced)	GATE_PASS_UNLOCKED
PRJ-2026-001	ABB Power Products Vadodara	4-Spindle High Speed Coil Winder	3 Wooden Crates	SIGNED (100% PASS)	AWAITING_BANK_UTR	COMMERCIAL_HOLD
PRJ-2026-003	Schneider Electric	Rotary Turret Indexing Station	2 Wooden Crates	PENDING_TRIAL_SIGN	PENDING_FINAL_PI	LOCKED_ZERO_SNAG

Shipping Dossier Verification Progress

- Pre-Delivery Inspection (PDI) Signed Cert — Verified
- Machine Operation & Maintenance Manual — Verified
- As-Built Electrical Schematics & Schematics — Verified
- Government E-Way Bill & Tax Invoice — Verified

3. MASTER DATA REQUIREMENTS & EXCEL INPUT SCHEMAS

Transporter & Carrier Master (transporters.xlsx)

Field Name (Excel Header)	Data Type	Mandatory?	Validation Rules / Constraints	Sample Industrial Value
`transporterCode`	String	Mandatory	Unique ID format TRN-XXX	**TRN-001**
`carrierName`	String	Mandatory	Registered logistics partner	**V-Trans India Ltd**
`gstin`	String	Mandatory	15-character GSTIN	**27AAACV4120K1Z8**
`vehicleTypes`	Dropdown	Mandatory	OPEN_TRUCK_17FT / CLOSED_CONTAINER_24FT	**CLOSED_CONTAINER_24FT**

Packing List & Crate Master (packing_lists.xlsx)

Field Name (Excel Header)	Data Type	Mandatory?	Validation Rules / Constraints	Sample Industrial Value
`boxNumber`	String	Mandatory	Crate number BOX-XX	**BOX-01**
`dimensionsMm`	String	Mandatory	L x W x H in millimeters	**2400 x 1800 x 2100**
`grossWeightKg`	Number	Mandatory	Total gross weight in KG	**2850**
`moistureProtection`	String	Mandatory	Packaging barrier method	**Vacuum VCI Bag + Silica Gel**

M10 After Sales, Warranty & Field Service

Route: `/after-sales` | Owner: **Head of After-Sales & Customer Service**

Post-commissioning asset history, 12-month client warranty tracking, spare parts dispatch, and preventive maintenance (PM) schedules.

1. VISUAL PROCESS FLOW CHART & STAGE GATES



2. MODULE DASHBOARD & REAL-TIME KPIS (WHAT PLANT USERS SEE)

After Sales, Warranty & Field Service Dashboard (Live Plant View) — Real-time sync active

ACTIVE FIELD TICKETS	MEAN TIME TO RESOLVE (MTTR)	WARRANTY MARGIN HEALTH	SCHEDULED PM COMPLIANCE
2 Open	**18.4 Hours**	**92.4%**	**98.0%**
SLA Under Control	SLA < 24 Hours	Within Warranty Budget	Preventive Care

Live Field Service Tickets & Support Queue

Ticket ID	Customer Company	Machine Asset ID	Reported Snag Issue	Ticket Priority	SLA Countdown	Assigned Engineer	Status
SRV-2026-024	ABB Power Products	AST-MHT-081	Wire breakage sensor false triggers	MAJOR	14 Hours Remaining	Suresh Jadhav	RESOLVED_CLOSED
SRV-2026-028	Schneider Electric	AST-MHT-076	Quarterly Preventive Maintenance Due	PREVENTIVE	5 Days Remaining	Ganesh Pawar	SCHEDULED_PM
SRV-2026-031	Lucas TVS Chennai	AST-MHT-084	HMI communication drop at start	CRITICAL	6 Hours Remaining	Nitin More	REMOTE_VPN_FIX

Installed Fleet Warranty Status

Machines Under Active OEM Warranty (12M)	18 Machines
Machines Under Annual Maintenance (AMC)	12 Machines
Machines with Expired Contracts	4 Machines
Customer Satisfaction Rating (CSAT)	4.9 / 5.0

3. MASTER DATA REQUIREMENTS & EXCEL INPUT SCHEMAS

Customer Asset Master (customer_assets.xlsx)

Field Name (Excel Header)	Data Type	Mandatory?	Validation Rules / Constraints	Sample Industrial Value
`assetId`	String	Mandatory	Unique ID format AST-MHT-XXX	**AST-MHT-081**
`projectId`	Dropdown	Mandatory	References Project Spine	**PRJ-2026-001**
`customerCode`	Dropdown	Mandatory	References Customer Master	**CUST-001 (ABB)**
`installationDate`	Date	Mandatory	Date of SAT sign-off	**2026-05-28**
`warrantyExpiryDate`	Date	Mandatory	Standard 12 months from SAT	**2027-05-27**

Preventive Maintenance (PM) Checklist (pm_schedules.xlsx)

Field Name (Excel Header)	Data Type	Mandatory?	Validation Rules / Constraints	Sample Industrial Value
`pmFrequency`	Dropdown	Mandatory	MONTHLY / QUARTERLY / HALF_YEARLY / YEARLY	**QUARTERLY**
`systemComponent`	String	Mandatory	Subsystem inspected	**Winding Spindle Bearings**
`maintenanceTask`	String	Mandatory	Procedure description	**Measure vibration levels and bearing temperature rise at 2000 RPM**

GUIDE

Departmental Data Collection & System Ingestion Protocol

Operational Readiness SOP

To ensure a flawless transition into live production on the Machine House (MH) ERP platform, all department heads must adhere strictly to the following 3-step master data submission protocol:

Step 1: Download Standard Excel Templates

Department leads will receive standardized .xlsx sheets matching the exact column names specified in Section 3 of this document. Columns must not be renamed, deleted, or reordered.

Step 2: Strict Data Hygiene Rules

Ensure zero duplicate codes across Item Codes (itemCode), Vendor Codes (vendorCode), and Customer Codes. Mandatory fields cannot be blank. Standard UOM tokens must be strictly used.

Step 3: Automated Dry-Run Validation

Upon uploading sheets into the MH ERP Admin/Import portal, the system runs an automated validation engine checking foreign keys and data types before committing records into MongoDB.

Department	Responsible Head	Required Excel Templates	Key Verification Checkpoint
IT / Administration	System Administrator	`company_profile.xlsx`, `users.xlsx`, `roles.xlsx`	Corporate tax IDs verified & plant IP whitelist assigned.
Sales / CRM	Business Development Head	`customers.xlsx`, `leads.xlsx`	GSTIN verified on GST portal; commercial payment terms baseline.
Design & Engineering	Design Lead	`bom_template.xlsx`, `drawings.xlsx`	3-Level BOM hierarchy verified; drawing files linked (.step/.pdf).
Purchase & Sourcing	Purchase Manager	`vendors.xlsx`, `item_specs.xlsx`	Standard lead days & MSME classification updated for suppliers.
Quality Assurance	Quality Lead	`qc_parameters.xlsx`, `defect_codes.xlsx`	Sampling plans & critical dimensional tolerances defined.
Store & Inventory	Store In-charge	`locations.xlsx`, `inventory_items.xlsx`	Physical rack/bin barcode mapping & initial stock verified.
Production & Assembly	Production Head	`work_centers.xlsx`, `assembly_stages.xlsx`	Bay capacities & standard assembly man-hours configured.
Dispatch & Logistics	Dispatch Executive	`transporters.xlsx`, `packing_lists.xlsx`	Carrier insurance verified; standard crate dimensions set.
After Sales Service	Customer Service Lead	`customer_assets.xlsx`, `pm_schedules.xlsx`	Client machine serial numbers & preventive maintenance frequency.